

Edge Detection using the Sobel Filter

Answer:

The 5×5 grayscale image patch:

$$\begin{bmatrix} 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \end{bmatrix}$$

Sobel-x Kernel.

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

1. Dimension of output feature map.

The equation used is $N - K + 1$.

where N = Size of image patch = 5

K = Size of Kernel = 3

$$\text{Dimensions} = 5 - 3 + 1 = \underline{\underline{3 \times 3}}$$

2. Convolution result

$$\text{Top left corner} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Sobel x kernel

$$= \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Element wise multiplication

$$\begin{aligned} &= (10 \times -1) + (10 \times 0) + (10 \times 1) + \\ &\quad (10 \times -2) + (10 \times 0) + (10 \times 2) + \\ &\quad (10 \times -1) + (10 \times 0) + (10 \times 1) \\ &= -10 + 0 + 10 - 20 + 0 + 20 - 10 + 0 + 10 \\ &= \underline{\underline{0}} \end{aligned}$$

3. Complete Output Matrix

$$\text{Position } (1,1) = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} &= \left\{ (-10 + 0 + 10) + (-20 + 0 + 20) \right. \\ &\quad \left. + (-10 + 0 + 10) \right\} \\ &= \underline{\underline{0}} \end{aligned}$$

$$\text{Position } (1,2) = \begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} &= (-10 + 0 + 80) + (-20 + 0 + 160) \\ &\quad + (-10 + 0 + 80) \\ &= 70 + 140 + 70 = \underline{\underline{280}} \end{aligned}$$

$$\text{Position (1, 3)} = \begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= (-10 + 0 + 80) + (-20 + 0 + 160) + (-10 + 0 + 80)$$

$$= \underline{\underline{280}}$$

$$\text{Position (2, 1)} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} = \underline{\underline{0}}$$

$$\text{Position (2, 2)} = \begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} = \underline{\underline{280}}$$

$$\text{Position (2, 3)} = \begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} = \underline{\underline{280}}$$

$$\text{Position (3, 1)} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} = \underline{\underline{0}}$$

$$\text{Position (3, 2)} = \begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} = \underline{\underline{280}}$$

$$\text{Position (3, 3)} = \begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} = \underline{\underline{280}}$$

$$\text{Final} \begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix}$$

4. Analysis

$$\text{Output} \begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix} *$$

location of edge = Column 3 ~~and 4~~

Why flat regions gives zero response?

It is because all neighbouring pixels have same intensity and hence zero response.

→ same intensity

$$\text{Eg: } \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= (-10 + 0 + 10) + (-20 + 0 + 20) + (-10 + 0 + 10)$$

$$= \underline{\underline{0}}$$